

**SIMATS ENGINEERING
ASSESSMENT TOOL 4**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Time Duration: 1 Hour

Total Marks:30

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I Athavan K(192524036)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Athavan k

Annexure A

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause ($\square$) if the goal is proved.

Question 2 – Student Assignment Submission Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

Question 3 – Library Membership Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member. **Goal**

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Question 4 – Placement Eligibility Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.

2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password. **Goal**

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause ($\square$).
5. Explain each resolution step and write the final conclusion.

1.

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Question 1 – Rain and Wet Ground

Task 1: Propositional Logic Representation

Symbols

- R = It rains
- W = The ground becomes wet

Knowledge Base (Propositional Form)

1. $R \rightarrow W$ (If it rains, the ground becomes wet)
2. R (It is raining)

Goal

Prove: W (The ground is wet)

Task 2: Conversion to CNF and Negation of Goal

Implication elimination: $R \rightarrow W \equiv \neg R \vee W$

Clause 1 (from premise 1): $\neg R \vee W$

Clause 2 (from premise 2): R

Negated Goal (added to KB): $\neg W$

Task 3: Step-by-Step Resolution

Step	Clauses Resolved	Resolvent
Step 1	R and $\neg R \vee W$	$\rightarrow W$
Step 2	W and $\neg W$	$\rightarrow \square$ (Empty Clause)

Task 4: Result

Since resolving the derived clause W with the negated goal $\neg W$ produces the empty clause ($\square$), the goal is PROVED by refutation. Hence, "The ground is wet" logically follows from the knowledge base.

Question 2 – Student Assignment Submission

Task 1: Propositional Logic Representation

Symbols

- S = A student submits the assignment
- M = The student receives internal marks

Knowledge Base (Propositional Form)

1. $S \rightarrow M$ (If a student submits the assignment, then the student receives internal marks)
2. S (Rahul submitted the assignment)

Goal

Prove: M (Rahul receives internal marks)

Task 2: Conversion to CNF and Negation of Goal

Implication elimination: $S \rightarrow M \equiv \neg S \vee M$

Clause 1 (from premise 1): $\neg S \vee M$

Clause 2 (from premise 2): S

Negated Goal (added to KB): $\neg M$

Task 3: Step-by-Step Resolution

Step	Clauses Resolved	Resolvent
Step 1	S and $\neg S \vee M$	$\rightarrow M$
Step 2	M and $\neg M$	$\rightarrow \square$ (Empty Clause)

Task 4: Result

The resolution of M with $\neg M$ yields the empty clause ($\square$), so the goal is PROVED. Therefore, it is logically valid to conclude that Rahul receives internal marks.

Question 3 – Library Membership

Task 1: Propositional Logic Representation

Symbols

- L = A person is a library member
- B = The person can borrow books

Knowledge Base (Propositional Form)

1. $L \rightarrow B$ (If a person is a library member, then the person can borrow books)
2. L (Priya is a library member)

Goal

Prove: B (Priya can borrow books)

Task 2: Conversion to CNF and Negation of Goal

Implication elimination: $L \rightarrow B \equiv \neg L \vee B$

Clause 1 (from premise 1): $\neg L \vee B$

Clause 2 (from premise 2): L

Negated Goal (added to KB): $\neg B$

Task 3: Step-by-Step Resolution

Step	Clauses Resolved	Resolvent
Step 1	L and $\neg L \vee B$	$\rightarrow B$
Step 2	B and $\neg B$	$\rightarrow \square$ (Empty Clause)

Task 4: Result

No further clauses can be generated after the empty clause is reached. The derivation of $\square$ PROVES the goal — Priya can borrow books.

Question 4 – Placement Eligibility

Task 1: Propositional Logic Representation

Symbols

- A = A student clears the aptitude test
- E = The student is eligible for placement

Knowledge Base (Propositional Form)

1. $A \rightarrow E$ (If a student clears the aptitude test, then the student is eligible for placement)
2. A (Arun cleared the aptitude test)

Goal

Prove: E (Arun is eligible for placement)

Task 2: Conversion to CNF and Negation of Goal

Implication elimination: $A \rightarrow E \equiv \neg A \vee E$

Clause 1 (from premise 1): $\neg A \vee E$

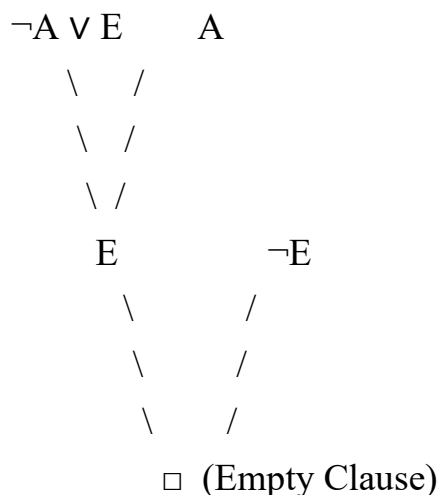
Clause 2 (from premise 2): A

Negated Goal (added to KB): $\neg E$

Task 3: Step-by-Step Resolution

Step	Clauses Resolved	Resolvent
Step 1	A and $\neg A \vee E$	$\rightarrow E$
Step 2	E and $\neg E$	$\rightarrow \square$ (Empty Clause)

Resolution Tree (Visual)



Task 4: Result

The empty clause ($\square$) is derived, so the goal is PROVED — Arun is eligible for placement.

Question 5 – Access Control System

Task 1: Propositional Logic Representation

Symbols

- P = The user enters the correct password
- Au = The user is authenticated
- Ac = The user is granted access

Knowledge Base (Propositional Form)

1. $P \rightarrow Au$ (If a user enters the correct password, then the user is authenticated)
2. $Au \rightarrow Ac$ (If a user is authenticated, then the user is granted access)
3. P (The user entered the correct password)

Goal

Prove: Ac (The user is granted access)

Task 2: Conversion to CNF and Negation of Goal

Implication elimination: $P \rightarrow Au \equiv \neg P \vee Au$

Implication elimination: $Au \rightarrow Ac \equiv \neg Au \vee Ac$

Clause 1: $\neg P \vee Au$

Clause 2: $\neg Au \vee Ac$

Clause 3: P

Negated Goal (added to KB): $\neg Ac$

Task 3: Step-by-Step Resolution

Step	Clauses Resolved	Resolvent
Step 1	P and $\neg P \vee Au$	$\rightarrow Au$
Step 2	Au and $\neg Au \vee Ac$	$\rightarrow Ac$
Step 3	Ac and $\neg Ac$	$\rightarrow \square$ (Empty Clause)

Explanation of Each Step

Step 1: Resolve P (fact) with $\neg P \vee Au$ (clause from rule 1) on complementary literals $P / \neg P \rightarrow$ derives Au ("the user is authenticated").

Step 2: Resolve Au (just derived) with $\neg Au \vee Ac$ (clause from rule 2) on complementary literals $Au / \neg Au \rightarrow$ derives Ac ("the user is granted access").

Step 3: Resolve Ac (just derived) with the negated goal $\neg Ac$ on complementary literals $Ac / \neg Ac \rightarrow$ derives the empty clause $\square$, confirming the refutation is complete.

Task 4: Result

Three resolution steps are needed because two chained implications must be traversed (password $\rightarrow$ authenticated $\rightarrow$ access). The empty clause ($\square$) is reached in Step 3, so the goal is **PROVED** — the user is granted access.